emergency care and medicine

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Brief Report

Large Language Models (LLM) for Emergency Department Triage Based on Vital Signs

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Abstract

Introduction: Large language models (LLMs) have proven effective in many different fields, including the allocation of scarce resources. Triage within emergency departments (ED) is a core process that ensures the sickest patients are seen in a timely manner. Relatively little research has examined the use of existing LLMs in the triage process. Methods: 12 widely available LLMs were provided with real-world patient triage vital sign data from an academic trauma center in a major metropolitan area. The LLMs were asked to assign a triage score to each patient based on this information alone. The deviation between each LLM triage score and the real-world triage score for each patient was calculated, and the absolute value of the deviation was calculated and then averaged across the entire dataset per LLM. The average absolute value of deviation (AAVD) could then be used to compare LLMs against each other. All LLMs were blinded to the real-world triage score and received no additional training or instruction. Results: The models with the highest concordance with real-world triage scores were Claude Sonnet 4.5 (AAVD: 0.37; 62.37% concordance), ChatGPT-5 Instant (AAVD: 0.39; 62.89% concordance), and Claude Opus 4.1 (AAVD: 0.40; 62.37% concordance). The least accurate models were Gemini 2.5 Flash (AAVD: 0.42; 43.81% concordance), ChatGPT-4o Mini (AAVD: 0.49; 45.36% concordance), and ChatGPT-o3 (AAVD: 0.48; 48.45% concordance). Conclusions: This study analyzes the ability of LLMs to triage emergency department patients based primarily on vital sign data. Certain LLMs demonstrated moderate concordance with real-world triage scores. LLMs may be able to synthesize objective vital sign data and provide a triage recommendation. Further study could involve clinical validation against patient outcomes.

Keywords: large language models; artificial intelligence in healthcare; emergency department triage; emergency severity index; clinical decision support; machine learning; emergency medicine; triage accuracy

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Academic Editors: George Notas and Zhongheng Zhang

Received: 30 January 2026
Revised: 20 February 2026
Accepted: 28 February 2026
Published: 5 March 2026

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1. Introduction

The use of large language models (LLMs) has significantly increased in recent years in medical and non-medical contexts [1,2]. LLMs are artificial intelligence (AI) machine learning models trained to detect patterns and trends, as well as converse with users using human language [3]. This ability to recognize trends and patterns holds particular relevance to medical situations, where these capabilities can inform clinical decision-making [4]. Certain LLMs have even been able to pass complex medical licensing exams, including the United States Medical Licensing Exam (USMLE), the Chinese National Nurse Licensing

Exam, and the German Medical State Examination [5]. One arena where AI and LLMs might hold particular relevance is within the emergency department (ED) triage process.

Medical triage refers to the prioritization of care based on a given patient's needs and the available resources [6]. Effective triage functions to ensure available resources are efficiently allocated [7]. There are multiple triage systems currently in use within emergency departments, including the Emergency Severity Index (ESI), the Manchester Triage System (MTS), and the Canadian Triage System (CTS) [8–10]. Delays in triage or inaccuracies in triage have been linked with adverse outcomes, highlighting a potential need for oversight within the triage process [11]. LLMs have shown early potential in the ability to triage ED patients, as well as efficiently allocate emergency medical services (EMS) resources [12–14].

In this study, we compared several LLMs for emergency department triage using a real-world dataset of emergency department patients. LLMs were blinded to the triage score each patient ultimately received. This allowed us to compare the concordance of various LLMs against the real-world, human-assigned triage score a given patient received. We sought to: (1) compare concordance and deviation between multiple LLMs and real-world triage scores and (2) identify whether models show systematic tendencies toward over- or under-triage.

2. Materials and Methods

2.1. Data Source

Real-world emergency department patient data came from an open-access deidentified dataset of ED patients from Beth Israel Deaconess Medical Center in Boston, MA, USA; specifically, the Medical Information Mart for Intensive Care (MIMIC), Version MIMIC-IV-ED Demo v2.2 [15,16]. Data were collected between 2011 and 2019, and the demo dataset contains 194 distinct patient encounters. Triage data included chief complaint, vital sign measurements (temperature, heart rate, respiratory rate, oxygen saturation, and blood pressure), and pain level (ranging from 1 to 10). Included with this basic triage data was the triage score originally assigned by the ED. This study used publicly available, deidentified data. As a secondary analysis of deidentified data, this study was exempt from institutional review board (IRB) review per institution policy.

2.2. Large Language Model (LLM) Use

A total of 12 different LLMs accessed on 9 October 2025 were used to triage the same 194 ED patients using solely the data points listed above. The 12 LLMs used were: Chat Generative Pre-trained Transformer (ChatGPT) (Versions 5 Instant, 5 Thinking Mini (Standard), 5 Thinking (Standard), o3. 4.1, 4o, and 4o Mini), Claude (Versions Opus 4.1 and Sonnet 4.5) (Anthropic, San Francisco, CA, USA), Grok (xAI, Palo Alto, CA, USA), and Gemini (Versions 2.5 Flash and 2.5 Pro).

To minimize the variability between each model's answer, the following prompt was given to each model along with the prespecified variables in an identical tabular format (chief complaint, vital signs, and pain level): "Patients show up in your ER, rank each patient (one per row) on ESI score 1–5." No further instructions were provided to the LLMs, and the aforementioned prompt was the only request given. Each patient was evaluated once per model, preventing any reruns. No application programming interfaces (API) were used.

2.3. Statistical Analysis

For each triage score determined by the 12 LLMs, the deviation from the real-world triage score assigned in the ED was calculated, and the absolute value of the deviation was calculated and then averaged across the entire dataset per LLM. The average absolute value

of deviation (AAVD) could then be used to compare LLMs against each other. The formula for AAVD is shown below in Figure 1, where j indicates the individual LLM, N represents the total number of patients, i represents the individual patient, LLMscore represents the triage score given by the LLM, and Pscore represents the human-assigned triage score given to each patient.

$$AAVD_j = \frac{1}{N}\sum_{i=1}^{N} |LLMscore_{ij} - Pscore_i|$$

Figure 1. Formula for average absolute value of deviation (AAVD).

Additionally, we calculated the concordance between each model and the real-world triage scores, as well as the rates of over-triage or under-triage for each model when compared to the real-world triage scores. The formula for concordance is shown in Figure 2, where j indicates the individual LLM, N represents the total number of patients, i represents the individual patient, LLMscore represents the triage score given by the LLM, and Pscore represents the real-world triage score given to each patient.

$$Concordance_j = \frac{\text{Number of patients where } LLMscore_{ij} = Pscore_i}{N} \times 100\%$$

Figure 2. Formula for concordance.

3. Results

The distribution of triage scores from each rater, including the provider and all LLMs, is listed as counts in Table 1. The AAVDs are shown in Table 2, presenting models Claude Sonnet 4.5, ChatGPT-4o, and ChatGPT-5 Instant with AAVDs under 0.40. The concordance between each of the twelve LLMs and the real-world triage scores is shown in Table 3. The models with the highest concordance with the human-assigned triage score were Claude Sonnet 4.5 (AAVD: 0.37; 62.37% concordance), ChatGPT-5 Instant (AAVD: 0.39; 62.89% concordance), and Claude Opus 4.1 (AAVD: 0.40; 62.37% concordance). The models with the lowest concordance were Gemini 2.5 Flash (AAVD: 0.42; 43.81% concordance), ChatGPT-4o Mini (AAVD: 0.49; 45.36% concordance), and ChatGPT-o3 (AAVD: 0.48; 48.45% concordance).

Table 1. Distribution of triage scores from the provider and each LLM for all patients.

Rater	Triage Score 1	Triage Score 2	Triage Score 3	Triage Score 4	Triage Score 5
Human	9	94	89	2	0
ChatGPT-5 Instant	13	96	75	8	2
ChatGPT-5 Thinking (Standard)	6	124	52	11	1
ChatGPT-5 Thinking Mini (Standard)	3	130	52	9	0
ChatGPT-4.1	16	114	64	0	0
ChatGPT-4o	0	94	85	15	0
ChatGPT-4o Mini	0	87	83	24	0
ChatGPT-o3	4	155	31	4	0
Claude Sonnet 4.5	3	92	94	5	0
Claude Opus 4.1	2	62	122	8	0
Gemini 2.5 Flash	10	61	109	14	0
Gemini 2.5 Pro	7	83	73	27	4
Grok	2	79	87	26	0

Table 2. The average absolute value of deviation (AAVD) between each LLM's triage score and the real-world human-assigned triage score across the entire dataset. The 95% confidence intervals (CIs) for each LLM are listed as well.

LLM	AAVD	Confidence Intervals (CIs)
Claude Sonnet 4.5	0.37	(0.29–0.44)
ChatGPT-4o	0.38	(0.30–0.46)
ChatGPT-5 Instant	0.39	(0.32–0.47)
Claude Opus 4.1	0.40	(0.33–0.48)
Gemini 2.5 Flash	0.42	(0.34–0.49)
ChatGPT-4.1	0.44	(0.37–0.52)
ChatGPT-5 Thinking (Standard)	0.47	(0.39–0.55)
ChatGPT-o3	0.48	(0.40–0.55)
ChatGPT-5 Thinking Mini (Standard)	0.48	(0.40–0.56)
ChatGPT-4o Mini	0.49	(0.40–0.58)
Grok	0.49	(0.41–0.58)
Gemini 2.5 Pro	0.60	(0.51–0.68)

Table 3. A list of each LLM used with rates of concordance between human-assigned triage score and LLM-assigned triage score, as well as rates of over-triage and under-triage across each LLM.

LLM	Concordance (%)	Over-Triaged (%)	Under-Triaged (%)
ChatGPT-5 Instant	62.89	19.59	17.53
Claude Opus 4.1	62.4	3.6	34
Claude Sonnet 4.5	62.37	6.7	30.93
ChatGPT-4o	55.15	8.76	36.08
Gemini 2.5 Pro	54.64	6.7	38.66
ChatGPT-5 Thinking Mini (Standard)	53.61	31.96	14.43
ChatGPT-5 Thinking (Standard)	53.09	30.41	16.5
ChatGPT-4.1	52.58	35.05	12.37
Grok	50.52	4.12	45.36
ChatGPT-o3	48.45	40.72	10.82
ChatGPT-4o Mini	45.36	12.37	42.27
Gemini 2.5 Flash	43.81	11.34	44.85

Models with the highest tendency to over-triage patients were ChatGPT-o3 (40.72% over-triaged), ChatGPT-4.1 (35.05% over-triaged), and ChatGPT-5 Thinking Mini (Standard) (31.96% over-triaged). Models with the highest tendency to under-triage patients were Grok (45.36% under-triaged), Gemini 2.5 Flash (44.85% under-triaged), and Gemini 2.5 Pro (38.66% under-triaged).

4. Discussion

4.1. Findings

In this study, we assessed various currently available LLMs in the task of ED triage by feeding them real-world ED patient data and compared the triage scores assigned by the LLM with the real-world, human-assigned triage scores each patient received. On average, LLMs deviated from the assigned triage score by 0.46 points and correctly triaged 53.79% of patients. Three individual LLMs (ChatGPT-5 Instant, Claude Opus 4.1, and Claude Sonnet 4.5) demonstrated over 60% concordance.

In this study, we had to assume the real-world triage score assigned to each patient was correct. However, this may not be the case: ED triage accuracy is a complex process influenced by patient characteristics, experience of the person performing triage, and multiple other factors [17,18]. ED triage accuracy should certainly be studied further.

Another major aim of our study was to determine whether certain models had tendencies to over-triage or under-triage relative to other models. Consistent differences between models may reflect differences in training and information processing. Differences in triage tendencies may mirror other model-specific characteristics that have been observed in other domains and with other tasks [19]. Another plausible explanation might be the ways each LLM conceptualizes triage and a specific degree of acuity. With triage scores, there are no precise definitions based on vital signs, but significant deviations from normal vital sign ranges are likely considered higher acuity [20]. Models might have specific thresholds for what constitutes a significant deviation. Namely, one model might automatically assign a triage score of 2 for a patient with a systolic blood pressure less than 90 mmHg, whereas a different model might consider a systolic blood pressure of 85 mmHg to be the threshold for significant hypotension warranting a higher acuity triage. These differences in a given model's underlying architecture likely contribute to the observed differences in triage tendencies. In terms of real-world applications to patients, tendencies of models to over-triage or under-triage could potentially have clinical effects. Based on our data, certain models had a tendency to under-triage or over-triage patients, whereas other models demonstrated more balanced triage tendencies. Models with low AAVD but high under-triage rates may present greater patient safety risks compared with models demonstrating balanced misclassification patterns. Our findings offer very preliminary data to suggest that LLMs may be able to assist triage in emergency departments as a potential decision-making aid, rather than a substitute for clinician judgment.

4.2. Limitations

Our study is associated with several limitations. First, the LLMs did not have the human aspect of being physically present with a patient. Rather, they were asked to triage patients based on only eight points of data. The clinician who initially triaged these patients had a wealth of additional information, largely based on their overall appearance and findings from an initial physical assessment, that the LLM did not have. Similarly, additional LLM information, such as access to electronic medical records, may be able to improve LLM triage accuracy. However, this limitation may actually mimic real-world practice, as one potential deployment of a language model for ED triage might involve providing triage score suggestions based on objective data, which could then take the AI recommendation into consideration. Similarly, we compared LLM scores to a single clinician's triage assignment without independent validation. Given that human triage has documented inter-rater variability, agreement with one clinician does not fully establish accuracy [21]. Mechanistically, we used a single prompt design without optimization. Prompt engineering might improve performance. Additionally, LLM versions change frequently, potentially limiting reproducibility.

Importantly, this study cannot determine whether LLM-assisted triage would improve clinical outcomes, reduce errors, or enhance operational efficiency. Such questions require prospective implementation studies with rigorous evaluation of patient safety and workflow impact. We also did not link triage scores to clinical outcomes, limiting our ability to assess whether LLM-human disagreements represented errors or appropriate judgment differences. Another limitation is the small sample size of this study. Our sample of 194 patients, while providing valuable proof-of-concept data, limits statistical power to detect differences between models and restricts generalizability. The particular dataset used for the study (MIMIC-IV-ED Demo v2.2) is publicly available, de-identified, and previously validated in prior publications but only contains a limited number of patients. We acknowledge that additional cases could alter effect size estimates and model ranking. Larger multicenter validation is needed. Further research could expand on this topic in a

variety of ways, including using more LLMs, asking each LLM to triage the set of patients on multiple occasions, using multiple triage systems, or increasing the number or diversity of patients analyzed (i.e., multicenter data).

5. Conclusions

This proof-of-concept study analyzes the ability of LLMs to triage emergency department patients. We provide a framework for the comparison of LLMs and human clinicians in assigning triage scores, namely, the AAVD metric. Certain LLMs can achieve moderate concordance (~62%) with real-world triage scores. These findings establish feasibility but do not yet conclusively determine clinical utility. Future studies must validate LLM triage performance against patient outcomes, assess safety in prospective trials, determine optimal workflow integration, and evaluate real-world implementation before widespread clinical deployment. Future work should assess how performance changes as LLMs are updated over time, including reproducibility under fixed prompts and inputs and the impact of version drift on concordance and triage tendency. Additionally, larger multicenter evaluations should be conducted to test generalizability and subgroup performance and to quantify model stability across different settings and populations.

Author Contributions: Conceptualization, A.C.S. and E.E.A.; methodology, T.G.L., W.C.H., A.C.S., D.J.A. and E.E.A.; software, T.G.L. and W.C.H.; formal analysis, T.G.L. and W.C.H.; investigation, T.G.L. and W.C.H.; resources, B.S.A. and A.C.S.; data curation, A.C.S.; writing—original draft preparation, T.G.L., W.C.H. and A.C.S.; writing—review and editing, A.C.S., L.A.A., E.E.A., B.S.A. and D.J.A.; visualization, T.G.L. and W.C.H.; supervision, A.C.S., B.S.A., D.J.A. and E.E.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: The authors declare no conflict of interest.

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